Lec27

12 March 2026 10:39 AM

Searching

Linear — Binary

arr = [1, 2, 3, 4, 5, 6] target = 6

[6, 2, 5, 3, 1] =4

6:46

6:00 — 7:00

6:30

6:45

6:52

6:45 — 6:52

48

45 — 48

46

[2, 3, 4, 5, 6, 7, 8, 9, 10] L=H target = 8 13

0 1 2 3 4 5 6 7 8

L H m

L = 0 5 7 8 9

H = 8

$\frac{0+8}{2} = 4$

$\frac{5+8}{2} = 6$

$\frac{7+8}{2} = 7$

$\frac{8+8}{2} = 8$

```
while(L <= H){
    mid = (L+H)/2
    if(arr[mid] == target){
        return mid
    }
    else if(arr[mid] > target){
        H = mid-1
    }
    else {
        L = mid+1
    }
}
return -1
```

$x^k <= n$

n = 149

k = 3

x = 1 2 3 4 5 6

$1^3 <= 149$

$2^3 <= 149$

$3^3 <= 149$

$4^3 <= 149$

$5^3 <= 149$

$6^3 <= 149$

216

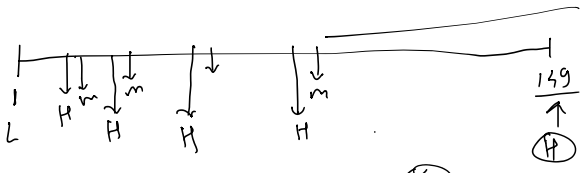
$6^3 = 216$

5 = x-1

```
x = 1
while(x^k <= n){
    x++
}
```

}

(x-1)



$75^3 <= 149$

$37^3 <= 149$

$18^3 <= 149$

$9^3 <= 149$

$4^3 <= 149$

$6^3 <= 149$

$5^3 <= 149$

ans = 5

$\frac{1+149}{2} = 75$

$\frac{1+74}{2} = 37$

$\frac{1+36}{2} = 18$

$\frac{1+17}{2} = 9$

$\frac{1+8}{2} = 4$

$\frac{5+8}{2} = 6$

$\frac{5+5}{2} = 5$

$\frac{6+5}{2} = 5$

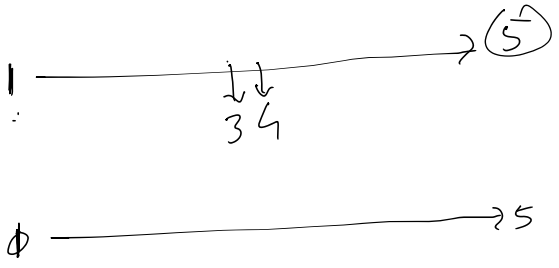
```
while(L<=H){
    mid = (L+H)/2
    if(mid^k <= n){
        ans = mid
        L = mid+1
    }else{
        H = mid-1
    }
}
```

tiktok

1, 2, 3, 4, 5, 6

1, 4

n = 5



$$\frac{1+5}{2} = 3$$

$$\frac{4+5}{2} = 4$$

$$\frac{0+5}{2} = 2$$